Averages from Tables

Exam code: 0607

Download PDF

Written by: Mark Curtis
Reviewed by: Dan Finlay
Updated on 23 January 2026

Averages from tables & charts

What are frequency tables?

- **Frequency tables** are used to **summarise** data in a **neat** format
  - They also put the data in **order**
- For example, the table below shows the number of pets in different houses along a street
  - The number of pets is the **data value**, ***x***
  - The number of houses is the **frequency**, ***f***
    - The **frequency** is **how many times** a data value is **recorded** (or **seen**)
- The **total frequency**, ***n***, can be calculated by **adding** together all the values in the **frequency column**

Number of pets (data *value*, *x*)	Number of houses (*frequency*, *f*)
0	2
1	7
2	6
3	4
4	1
	Total frequency (n) = 20

How do I find the mode from a frequency table?

- The **mode** is the **data value** with the **highest frequency**
  - The mode for the example above is 1 pet per house
    - The mode is **not** the frequency, 7, this is the number of houses that have exactly 1 pet

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- The **median** is the **data value** in the **middle** of the **frequency**
  - It is the $\left(\frac{n+1}{2}\right)^{th}$ value, where n is the total frequency
- From above, $n = 20$ so the median is the $\left(\frac{20+1}{2}\right)^{th}$ = 10.5^{th} value in the table which is the mean of the 10th and 11th values
  - The first two rows have a **combined** (**cumulative**) frequency of 2 + 7 = 9
  - The first three rows have a combined frequency of 2 + 7 + 6 = 15
  - Therefore the 10th and 11th values are in the third row (x = 2)
    - The median is 2 pets per house

How do I find the mean from a frequency table?

- The **mean** from a **frequency table** has the following **formula**:
  - $\text{mean} = \frac{\text{total of 'data value} \times \text{frequency'}}{\text{total frequency}}$
    - It helps to create a new column of 'data value × frequency'
    - Add up the values in this column
    - Divide by the total frequency
- The mean is $\frac{35}{20}$ = 1.75 pets per house
  - Means do not need to be whole numbers

Number of pets (data *value*, *x*)	Number of houses (frequency, *f*)	data value × frequency (*xf*)
0	2	0 × 2 = 0
1	7	1 × 7 = 7
2	6	2 × 6 = 12
3	4	3 × 4 = 12
4	1	4 × 1 = 4
	Total = 20	**Total = 35**

How do I find the range from frequency tables?

- The **range** is the **difference** between the largest and smallest **data values**
  - The range above is 4 - 0 = 4

What else should I know about frequency tables?

- **Tables** can be **converted** back into a **list of data values** using their **frequencies**
  - From above, 0 pets were recorded twice, 1 pet was recorded 7 times, 2 pets were recorded 6 times, etc
    - The list of pets recorded is 0,0,1,1,1,1,1,1,1,1,2,2,2,2,2,2,3,3,3,3,4
- You could then find the **mode**, **median** and **mean** from this list of numbers

Worked Example

The table shows data for the shoe sizes of pupils in class 11A.

Shoe size	Frequency
6	1
6.5	1
7	3
7.5	2
8	4
9	6
10	11
11	2
12	1

(a) Find the mean shoe size for the class, giving your answer to 3 significant figures.

Answer:

It helps to label shoe size (x) and frequency (f)
Add an extra column and calculate the values of 'shoe size × frequency', (xf)
Find the total frequency and total xf value

Shoe size (x)	Frequency (f)	xf
6	1	$6 \times 1 = 6$
6.5	1	$6.5 \times 1 = 6.5$
7	3	$7 \times 3 = 21$
7.5	2	$7.5 \times 2 = 15$
8	4	$8 \times 4 = 32$
9	6	$9 \times 6 = 54$

12	1	12 × 1 = 12
	Total = 31	Total = 278.5

Use the formula that the mean is the total of the xf column divided by the total frequency

$$\text{Mean} = \frac{278.5}{31} = 8.983\,870\ldots$$

Give your final answer to 3 significant figures

The mean shoe size is 8.98 (3 s.f.)

Note that the mean does not have to be an actual shoe size

(b) Find the median shoe size.

Answer:

The median is the $\left(\frac{n+1}{2}\right)^{th}$ value where n is the total frequency

$$\frac{n+1}{2} = \frac{31+1}{2} = \frac{32}{2} = 16$$

The median is the 16th value
There are 1 + 1 + 3 + 2 + 4 = 11 values in the first five rows of the table
There are 11 + 6 = 17 values in the first six rows of the table
Therefore the 16th value must be in the sixth row

The median shoe size is 9

(c) Find the range of the shoe sizes.

Answer:

The range is the highest shoe size minus the lowest show size

12 - 6

The range of the shoe sizes is 6

Teacher only

Test your students on this in just a few clicks

- Create exam-aligned tests
- Tailor format & difficulty
- Quizzes or homework in minutes

Try Test Builder →

Test yourself

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.

Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.